

AWI-ESM3 ↔ PISM orography and SMB coupling

design notes and decisions

Companion to the moving-cavity investigation report. Same convention: superseded claims are struck through, not deleted.

Last updated: 2026-08-03 (second revision)

Abstract

The moving-cavity campaign closed the *ocean* side of the AWI-ESM3 ↔ PISM cycle: FESOM cavity melt reaches PISM, and PISM geometry moves the FESOM mesh, the OIFS land-sea mask and the WAM sea points. The *atmosphere* side is still open in both directions. PISM's surface mass balance is not driven by our OIFS, and PISM's evolving surface elevation does not reach OIFS. This document records what `develop-is` actually has today, what the EC-Earth4 reference design does, and what we decided to build on branch `awiesm3-is-orography-and-smb-coupling`.

Contents

1	Goal	1
2	What <code>develop-is</code> has today	2
2.1	The atmosphere forcing PISM has been eating is a stub	2
2.2	The orography return path exists in the binary but is switched off	2
2.3	Known-good pieces we inherit	2
3	The EC-Earth4 reference design	2
4	Decisions	3
4.1	Use the ISM-mapper, as EC-Earth does	3
4.2	Both SMB schemes, dEBM as the default	3
4.3	SMB first, orography second	3
4.4	Feature switch	4
5	Built so far	4
5.1	The ISM-mapper is an <code>esm_tools</code> component	4
5.2	The PISM grid is in <code>ocp-tool</code>	4
5.3	How the weights are actually made	5
5.4	<code>feom</code> no longer written by default	5
5.5	OIFS tree current	5
6	Open items	5
7	Provenance	6

1 Goal

Close the atmosphere half of the cycle, in both directions:

- **SMB (OIFS → PISM).** PISM’s surface forcing comes from the OIFS leg that just ran, not from a canned file.
- **Orography (PISM → OIFS).** The ice-sheet surface elevation OIFS runs with follows the ice sheet PISM produced, so the atmosphere sees the ice sheet it is forcing.

2 What `develop-is` has today

2.1 The atmosphere forcing PISM has been eating is a stub

`couplings/oifs/just_copy_stuff.functions` copies `atmosphere_file_for_ice.nc` verbatim out of `a270122/lars_an`. Confirmed in `eval5`: the staged file is a 192×96 ECHAM Gaussian grid carrying `temp2`, `tsurf`, `aprt`, `orog`, `swd`, `cc`, `q2m`, `tau`, `emiss`, `uv`, `T0Aswd`.

Taint. Every PISM leg of the campaign, `eval5` and `fgdbg02` included, ran dEBM on a 130 ka paleo ECHAM atmosphere. Our own OIFS never reached the ice sheet. This is a *surface* taint only: it does not touch the FESOM-side melt diagnosis or the ocean-side results, which are forced from FESOM output directly. It does mean no PISM surface mass balance number from this campaign is a statement about AWI-ESM3.

2.2 The orography return path exists in the binary but is switched off

`suorog.F90:151-250` in the `develop-is` OIFS build already implements both halves:

- `ECE_CPL_ISMM` reads `usurf_oifs` from `{prefix}_pism2ece.nc`, multiplies by `RG`, patches `ZZOROG` where `usurf > 0`, and restores spectral consistency through `REESPE/SPEREE`.
- `ECE_LANDICE` reads `plit_oifs` from the same file and publishes it through `SURFACE_SET_LANDICE_LOCAL`.

We run `ECE_LANDICE` only. And the `plit` we write is not PISM’s ice mask: `couplings/fesom/make_plit_oifs.py` derives it from the regenerated ICMGG land–sea mask capped at 60°S . So no PISM geometry reaches the atmosphere today, in either field.

Verified in `eval5/couple/antar_pism2ece.nc`: dimensions `ny=1`, `nx=40320` (TCO95 octahedral), one variable, `plit_oifs`. No `usurf_oifs`.

2.3 Known-good pieces we inherit

- The -HUGE cold-start hazard from the EC-Earth notes is already fixed in our OIFS fork: `cplng2_data_mod.F90:354` initialises `OASIS_OUT` fields to `0.0_JPRB`.
- dEBM is wired and running. `eval5_pism.yaml` sets `ablation_method: DEBM`, and the couple dir carries `dEBMmain` plus its `SMB/SNH/A/ tsurf` output per decade. Only its *input* is wrong.
- The per-leg feedback-file staging already works. `antar_pism2ece.nc` is staged into `work/` through `add_input_in_work`, so it no longer depends on a file surviving in `work/` by luck. Adding a second variable to that file needs no new plumbing.

3 The EC-Earth4 reference design

Recorded from `ece4-pism-coupling.md`. Two tiers at different times:

1. **In-line OASIS component, the ISM-mapper** (`sources/ism-mapper`, OASIS name ISMMAP).
Runs as an MPI rank alongside OIFS for the whole leg. Receives precip, evaporation, runoff, soil temperature and SST from OIFS every coupling step, accumulates, and writes `{region}_smb_{year}.nc` monthly. In the other direction it reads the previous PISM feedback file and sends `plit` to OIFS and basal melt plus discharge to the runoff-mapper.
2. **Offline PISM driver** (`coupling_ece4_pism_v3.py`), run between legs. Turns the accumulated SMB file plus ocean output plus current PISM geometry into PISM forcing, runs PISM for the year, and builds `{region}_pism2ece_{year}.nc`, which the *next* leg's suorog and ISM-mapper consume.

Tier 2 is what our iterative coupling already is. Tier 1 is what we do not have.

4 Decisions

4.1 Use the ISM-mapper, as EC-Earth does

Decided. Port `sources/ism-mapper` into the `develop-is` build and add it as an OASIS component, rather than synthesising the SMB forcing from XIOS output between legs.

The considered alternative. Everything dEBM needs is already in our XIOS output. The monthly remapped stream carries `2t`, `skt`, `ssrd`, `strd`, `tisr`, `tcc`, `2d`, `msl`, `10u`, `10v`, `lsp`, `cp`, `sf`, `e`, `ro`, and the de-accumulation pattern is already in the tree: `field_def_ljpg_safe.xml.j2` divides by `{{ oifs.flux_accumulation_period }}`, including `1000*tp/...` for precipitation and `sqrt(ewss*ewss + nsss*nsss)/...` for stress. Arithmetic checked against `eval5` year 1903: `ssrd` = 6.845×10^5 and `tisr` = $1.226 \times 10^6 \text{ J m}^{-2}$ per one-hour step give 190 and 340 W m^{-2} , and `lsp+cp` = $1.13 \times 10^{-4} \text{ m/step}$ gives $3.14 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$, all correct global means. So a jinja-gated `file_def` block would have delivered the forward direction with no new component and no rebuild.

Why the ISM-mapper anyway. XIOS solves the *forward* direction only. It cannot deliver `plit` into the live atmosphere, and it cannot route PISM basal melt and discharge into the ocean as freshwater. Both of those need a component inside the OASIS exchange. Since one has to exist, having two mechanisms for one coupling is worse than having one. Staying on the EC-Earth component also keeps CISSEMBEL available as a future surface scheme and keeps this work mergeable with EC-Earth4 developments rather than diverging from them.

Cost, corrected twice. The original entry read: ~~AWI-ESM3 does not use `rdy2epl`, so adding a PISM regional grid means teaching the per-leg regeneration to emit a `pism-antar` grid with SCRIP weights to and from TCO95 on every cycle. This is the expensive part of the port and the main schedule risk. That overstated it, but a first correction (OASIS computes the weights itself at runtime, and the PISM grid does not move, so there is no per-cycle work)~~ was also wrong. What is actually true is §5.3: weights are *precomputed* by `ocp_tool/oasis_weights.py` through pyOASIS, and that machinery is general, so PISM costs a few `Link` entries rather than new code. But the links do still regenerate every mesh-change leg, because the OIFS land-sea mask moves and it moves exactly where PISM is.

4.2 Both SMB schemes, dEBM as the default

Wire the forcing so it can feed either dEBM (insolation-driven melt, already running) or the EC-Earth direct path, $\text{climatic_mass_balance} = P_\ell + P_s - E - R$ with `ice_surface_temp` from soil

layer 4 merged with SST. Selected by a runscript switch, defaulting to dEBM. The direct path costs only `e` and `ro` in the field set, so it is cheap to carry from the start, and having both lets the two be compared on identical legs.

4.3 SMB first, orography second

The SMB half is testable without an OIFS rebuild once the component is in place; the orography half needs `ECE_CPL_ISMM` on and a rebuild. Land and validate SMB against a leg first.

4.4 Feature switch

`smb_coupled`, defaulting true for `develop-is` and false elsewhere, in the same spirit as the existing `general.interactive_mesh` flag. All new wiring sits behind it so that `develop` and `develop-cc` are untouched.

5 Built so far

5.1 The ISM-mapper is an `esm_tools` component

Not a hand-built binary: `configs/components/ismm/ismm.yaml`, in the `develop-is` coupling, always built, with `smb_coupled` deciding at runtime whether it joins the launch. Notes worth keeping:

- It is not a standalone repository. It lives inside `ecearth4` under `sources/ism-mapper`, so the component clones `ecearth4` and builds that subtree. The clone is 37 MB, not the 663 MB the local `.git` suggests: the local copy carries three remotes and all of upstream, and every `sources/*` model is a submodule that is not recursed into. We get none of EC-Earth's NEMO, OIFS, XIOS or PISM, which is what we want. If recursive cloning is ever switched on, that changes.
- The branch is required, not preferred. `cissembel-levante-intel` carries 5799a2a, which fixes an `EBMfinalize` use-after-free that crashes any run with a non-CISSEMBEL region. That is our path.
- CISSEMBEL is a *link-time* dependency even on the direct path: `ism-mapper_run_mod.F90:7` has an unguarded `use modEBMworld` and the link line ends in `libebm_cpl.a`. At runtime `EBMworld` is only called for regions with `ISMRGNCIS(n)=.true.`, so it never executes for us. It is built from the same clone, not its own upstream, so the two cannot drift.
- The committed `Makefile` carries ECMWF HPC2020 paths and `mpifort`. Everything it needs is a plain make variable and command-line macros beat file assignments, so the machine configuration is passed in `comp_command` rather than by patching a tracked file. `OASIS_LIBS` differs: ours is `libpsmile.a`, EC-Earth's is `libpsmile.MPI1.a`.
- One real upstream bug fixed (41e0d8e): `CISSEMBEL/src/makefile` listed `mod_domain.F08` in `MOD_SRC0` and no such file exists in the tree, so a clean clone failed to build on any machine.

Verified. Built end to end from a fresh clone. The open risk was whether code written against OASIS 5.2 links against our 5.0-smhi; it does, with no source change. The binary carries `oasis_init_comp`, `oasis_def_var`, `oasis_enddef`, `ebmworld` and `ebminitialize`.

5.2 The PISM grid is in ocp-tool

The ISM-mapper is an OASIS component, so the PISM domain has to appear in `grids.nc/masks.nc/areas.nc`. This went into **ocp-tool**, not `esm_tools`: `ocp-tool` is already the tool sitting between PISM and AWI-ESM3, it is where the weight generation lives (§5.3, itself ported from `rdy2cpl`, whose `PismGrid` class EC-Earth uses for exactly this grid), and `ocp_tool/grids/` already has the abstraction, a class with `cell_latitudes`, `cell_longitudes`, `cell_corners`, `cell_areas`, `cell_masks`. Putting it there also deleted a planned step: `ocp-tool` writes `ismp` into the OASIS files itself, so nothing has to append fragments afterwards or keep them in sync.

Centres come from the PISM file's own `lat/lon`. Corners are inverse-projected from the cell corner (x, y) . Areas are spherical excess over two triangles, not a plane formula: the domain wraps the pole, so cells on the ± 180 meridian would get a longitude span of $\sim 2\pi$.

Verified on `antarct_cr` 8 km, 761×761 . Total area 36.03 against 36.28×10^6 km² from the plane area corrected by the polar stereographic scale factor, -0.70% . Cell areas run 67.0 km² at the pole to 53.8 at the outer corner against a nominal 64, the right sign and size for `lat_ts = -71`. All 579121 cells wind counterclockwise seen from outside the sphere, which SCRIP requires and the area check cannot detect. A full TCO95/CORE3 run puts `ismp` in all three files, bit-identical to an independently generated reference.

5.3 How the weights are actually made

Worth writing down, because it was easy to get wrong. Weights are *not* left to OASIS at model runtime. `ocp_tool/oasis_weights.py` precomputes them with `pyOASIS`, taken from the coupled model's own OASIS build (compiled `OASIS_WITH_PYOASIS=ON`) rather than a second install. The design is ported from `rdy2cpl`: run the coupler itself with one MPI rank per unique coupling link, let each rank define its source and target grids and a variable pair, and call `enddef()` so OASIS/SCRIP writes that link's `rmp*.nc`.

Two consequences for us. First, grid geometry is read back from the very `grids.nc/masks.nc/areas.nc` that `ocp-tool` writes, so the `ismp` entry of §5.2 is exactly what a PISM link consumes; nothing recomputes geometry. Second, this needs compute nodes: `esm_tools` drives it with `OCF_WEIGHTGEN_DRIVER_PY`, `OCF_WEIGHTGEN_WORKER_PY`, `OCF_WEIGHTGEN_THREADS` (64) and `OCF_WEIGHTGEN_PARTITION` (`compute`), and it is the one-rank-per-link parallelism that makes it tractable at all.

The link set is declared per setup, currently `awiesm3_feom_links()`, five `Link` entries carrying the transformation strings verbatim ("`GAUSWGT D SCALAR LATITUDE 1 25 0.1`" and friends). PISM needs the same treatment: `atmosphere→ismp` for the forcing fields and `ismp→atmosphere` for `plit`.

5.4 feom no longer written by default

Found on the way: `ocp-tool` was appending the FESOM grid to the OASIS files unconditionally. FESOM writes its own `feom` entry at runtime in domain-decomposition order, and what `ocp-tool` writes is mesh order, so the two disagree; `couplings/fesom/permute_feom_to_runtime.py` exists because of that ordering difference. Now behind `ocean.write_oasis_grid`, default off. *This changes default output for every FESOM config, not just ours.*

5.5 OIFS tree current

Merged `vendor/main` (14fa22d), which had not moved since 2026-07-06, and cherry-picked Violet's `ECE_ISMM_SET_STATE` refactor (5779a3d) so the ISM field fill is in its own file rather than inline in `callpar.F90`, which is the code the SMB work extends. Added the missing `intfb` include for it (38c3ab0): the call passes six derived types and had no explicit interface, so a signature change

would have gone uncaught. Jorge’s ice-sheet coupling is already in vendor main ([a2e46d5](#)), and our suorog.F90 is byte-identical to it, so the orography read we plan to use is upstream code.

6 Open items

- **OASIS field set for the direct path.** ecearth.F90:360 declares A_Evap_ISM, A_SST_ISM, A_Soil4T_ISM and PLIT_<region>; cnt4.F90:1336 issues the ISM_SND exchange. The grid now exists (§5.2); what is still missing is the declaration in awiesm3.yaml, the ismp grid entry plus ismm.coupling_fields and the atma->ismp/ismp->atma directions with a coupling method. Until that lands ECE_CPL_ISMM must stay off, because the fields would be declared with no partner.
- **PISM links for the weight generator.** Add the atmosphere↔ismp Link entries next to awiesm3_feom_links() (§5.3). They regenerate every mesh-change leg along with the feom ones, because the OIFS mask moves where PISM is; the ismp geometry itself never changes.
- **ismm in the launch.** It has to be an MPI rank whenever it runs, or OASIS enddef fails. That is what smb_coupled has to do, and the switch itself does not exist yet: it is named in comments but nothing reads it.
- **Per-leg namelist.ismm** (RunLengthSec, TimeStepSec, StartDate, the region lists) and the cpl_dir/{aux,prev_ staging.
- **Consume the SMB.** Turn {region}_smb_{year}.nc into atmosphere_file_for_ice.nc plus its names file, replacing just_copy and the 130 ka stub of §2.1.
- **Push before a clean build works.** The component clones remotes, not working copies, so esm_tools, oifs-48r1 and ecearth4 all have to be pushed before a fresh esm_master comp-awiesm3-develop-is succeeds for anyone, including us.
- **atlas 0.31.1 → 0.39.0** came in with the vendor merge and has not been built yet.
- **Pre-existing ocp-tool bug, left alone.** _append_fesom_grid_to_oasis_files guards its optional import with except ImportError, but import pyfesom2 raises AttributeError when cmocean meets a matplotlib that has dropped matplotlib.cm.register_cmap, so the whole run dies. Widening the catch would be worse: it would silently ship OASIS files with no ocean grid. Now moot on the default path, since that function is no longer called unless asked for.
- **orog for the downscaling reference.** dEBM needs the low-resolution elevation OIFS actually ran with. field_def.xml:176 has a z, but it is in the 3D_ml group, so it is model-level geopotential, not surface. Source it from the ICMGG init file we already stage and regenerate per leg, read with eccodes for param 129, mirroring what make_plit_oifs.py already does for lsm. Once the orography feedback is live this tracks the ice sheet automatically.
- **usurf_oifs writer.** PISM usurf on the 8 km polar-stereographic grid must be conservatively remapped into native OIFS gridpoint order (40320 for TCO95) and written alongside plit_oifs into antar_pism2ece.nc. EC-Earth uses precomputed pism2ifs_con_weights_{region}.nc; ours must be regenerated per leg with the rest.
- **Real plit.** Once PISM geometry is available on the OIFS grid, replace the ICMGG-derived plit of §2 with PISM’s own ice mask.
- **Freshwater return.** PISM basal melt and discharge currently do not reach FESOM at all. Decide whether to re-target the runoff-mapper’s ocean side to FESOM or wire the ISM-mapper SND_OCE fields into FESOM’s surface freshwater coupling. Not needed for the SMB or orography halves, but it is the remaining leg of the mass budget.

- `flux_accumulation_period` is defined only in `oifsamip.yaml:90`. If any XIOS-side derived field is used for validation it needs a default for `awiesm3`.

7 Provenance

Reference document `esm_tools/ece4-pism-coupling.md`, written from the EC-Earth4 checkout at `/work/ab0246/a270092/model_codes/ecearth4`. Runtime evidence from `/work/ab0246/a270092/runtime/awiesm3-d`

repository	branch	commits	from
<code>esm_tools</code>	<code>awiesm3-is-orography-and-smb-coupling</code>	<code>55a79350f</code> , <code>da8c1e069</code>	<code>c327e5dc1</code>
<code>ocp-tool</code>	<code>feat/pism-oasis-grid</code>	<code>e29f59a</code>	<code>7e76751</code> (master)
<code>oifs-48r1</code>	<code>movcav-landice+co2-concdriven</code>	<code>14fa22d</code> , <code>5779a3d</code> , <code>38c3ab0</code>	<code>45a7bea</code>
<code>ecearth4</code>	<code>cissembel-levante-intel</code>	<code>41e0d8e</code> (pushed)	<code>5cc50a0</code>

`da8c1e069` is the standalone PISM-grid script that preceded the `ocp-tool` port. It is superseded by `e29f59a` and should be removed once the `ocp-tool` branch merges; its output, in `/work/ab0246/a270092/input/pism/` is what `ocp-tool`'s is diffed against.